

Sweet Memories

Agent001 recounts his recent memory shopping spree

Motherboards and Processors this month! There's another component that works as hard as your processor most of the time—and is even more important as a limiting factor a.k.a. bottleneck to performance. Yes, I'm talking about good old Random Access Memory.

There's only one real memory standard today and that's DDR2. DDR is officially dead, and nobody is using it for new PCs, not even value PCs. DDR2 had initial teething issues with performance—mainly due to higher latencies, but all those have been ironed out, and the sheer bandwidth DDR2 offers over DDR is significant. DDR is still available in the market—but it's mostly older stock, since manufacturers have stopped making it. As a result, DDR memory is very expensive now—and those who need it for older systems will face the simple market reality of demand and supply. I found DDR memory at prices roughly twice that of DDR2 in Lamington Road, and the prices fluctuated from dealer to dealer—although many dealers flatly stated they didn't stock DDR any more.

DDR2 is available in mainly two speed options—DDR2 667 and DDR2 800 MHz frequencies. The higher frequencies (1000 and 1066 MHz) are very expensive, and do not justify the premium. What I came across in shops was mostly value RAM. This is basically RAM based on cheaper memory chips, the characteristic being slower speeds and higher latencies. Enthusiast memory will have higher speeds at lower latencies but typically requires higher voltages to operate (1.8V is the default for DDR2), which means your board should support moderate overvoltage.

Generally, higher-performance (read enthusiast) memory—which performs better with demanding applications—typically has heat spreaders on the memory strip. I must caution all our readers: some manufacturers are resorting to the cheap tactic of putting attractive aluminium heatsinks on cheap value memory, and vendors call this “high performance” memory, and you'll subsequently be charged 500 to a thousand bucks more for nothing more than eye-candy! Look at the latency figures to base a purchase decision on, and not attractive heatsinks. If you're buying DDR2 667 MHz, look for latency figures of 4-4-4-12. For DDR2 800, timings should be 5-5-5-15 at the most; anything more and your memory is going to be slow irrespective of the frequency. For

DDR 400, 3-3-3-8 should suffice. (Lower timings are better.)

I nowadays recommend 2 GB of memory. Two reasons: your PC will speed up in general, and Vista—which is a memory hog. And you don't want to bottleneck a good processor and decent graphics card, do you? RAM is the expressway in your PC and handles data traffic—more is always good, although as of now you won't see any significant gains with more than 3 GB, and 2 GB is the sweet spot for Windows XP.

I was searching for a 2 GB kit for a friend who's upgrading from a 1 GB kit. The most common brands for DDR2 memory available are Transcend and Kingston. 1 GB of Kingston DDR2 667 MHz memory costs Rs 1,450, and Transcend is some 100 bucks more. Both manufacturers offer “lifetime” warranties, which equates to five years in the case of both vendors. This is a three-year replacement warranty plus a two year repair warranty. A 2 GB kit will cost you around 2,750 bucks (Transcend has a 2 x 1 GB kit). The slightly lower price for a 2 GB kit means you're getting more value for money for buying more memory! Corsair value RAM is also available at Rs 1,700 per GB—dual channel kits are just shy of the 3,500-rupee mark. Dynet is one of the locally-available brands, and is available for as little as Rs 1,300 per GB. Digital, another local brand retails at Rs 1,250 per GB—terrific value for money, but I'd rather pay 300 bucks more for blissful peace of mind.

DDR2 800 MHz is a little more expensive, but worth it if you're going for faster components. 1 GB of DDR2 800 RAM will cost you around 2,000 bucks—both Transcend and Kingston are available for nearly the same price. What's worth a mention is, the price fluctuates as well—for example, 1 GB of Transcend DDR667 was Rs 1,650 last week, this week it was 1,450. It could go up next week!

DDR3 is available, too, but prices are so restrictive at the moment that you're better off with faster DDR2. I don't expect DDR3 to make big inroads into DDR2, at least for another year. Only one big vendor had stocks of DDR3 from Corsair, and the 2 GB dual-channel kit (two sticks of 1600 MHz memory) was priced at Rs 23,000. A serious premium which I think is totally uncalled for.

I ended up buying a 2 GB Transcend kit (DDR2 667) for 2,750 bucks—a superb deal for my friend who had paid 3,500 bucks last year for his 1 GB kit! ☺

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